5.3 Graphing Radical Functions

Name _____ Period ____ Date _____

Algebra II 5.3 · Student sheet

I can…

- graph radical functions.
- find the domain and range of a radical function.
- describe and write transformations of radical functions.

Vocabulary · fill in the missing word

Radical function — a function with the _____ in a radicand, like $f(x) = \sqrt{x}$ or $f(x) = \sqrt[3]{x}$

Domain of a square root function — the _____ that make the radicand ≥ 0

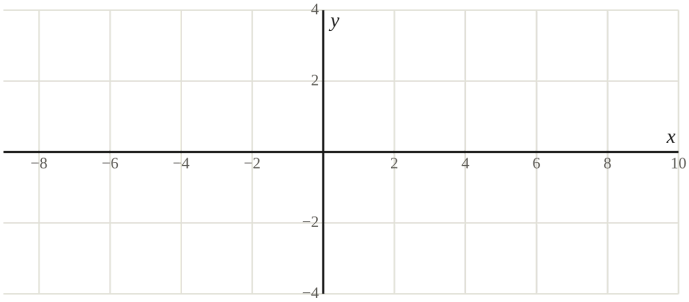
Transformations — $g(x) = a \cdot \sqrt[n]{x - h} + k$: a _____/shrinks (and reflects if $a < 0$), h moves left/right, k moves up/down

Part 1 · The parent functions

Complete each table. Then graph both functions.

x	0	1	4	9
$\sqrt{x}$				

x	−8	−1	0	1	8
$\sqrt[3]{x}$					



Desmos. Type $y = a\sqrt{(x - h)} + k$ and add sliders. What does each number do?

a: _____

h: _____

k: _____

Part 2 · Notes: fill in the blanks

$f(x) = \sqrt{x}$ has domain _____ and range _____. $f(x) = \sqrt[3]{x}$ has domain _____ and range _____.

In $g(x) = a \cdot \sqrt[n]{(x - h)} + k$: a stretches or shrinks the graph, and $a < 0$ _____ it in the x-axis.

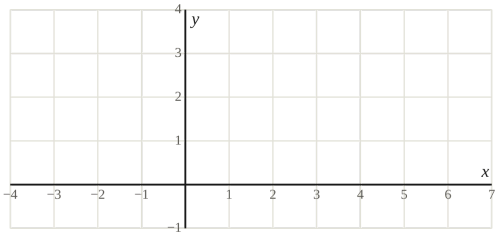
h translates the graph _____ ($h > 0$) or _____ ($h < 0$). k translates it _____ ($k > 0$) or _____ ($k < 0$).

To find the domain of a square root function, set the radicand _____.

Part 3 · Examples (work them with the class)

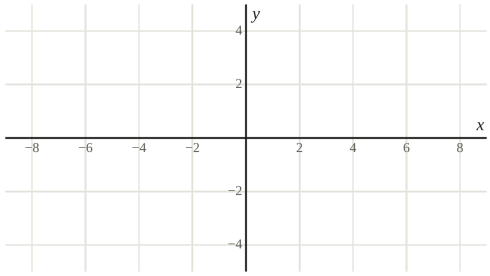
Example 1

Graph $f(x) = \sqrt{x + 3}$. Identify the domain and range.



Example 2

Graph $g(x) = -2\sqrt[3]{x}$. Identify the domain and range.



Example 3

Describe the transformation of $f(x) = \sqrt{x}$ represented by $g(x) = -\sqrt{x - 4} + 1$.

Example 4

Let the graph of g be a vertical stretch by a factor of 3, followed by a translation 2 units left, of the graph of $f(x) = \sqrt[3]{x}$. Write a rule for g.

Example 5

Find the domain and range of $f(x) = \sqrt{2x - 6} + 1$.

Example 6

After a car brakes, the speed s (in miles per hour) it was going can be estimated from the length d (in feet) of its skid marks by $s = \sqrt{22.5d}$. How fast was a car going if its skid marks are 40 feet long?

Part 4 · Partner practice

Partner 1

Describe the transformation of $f(x) = \sqrt[3]{x}$ represented by $g(x) = \frac{1}{2}\sqrt[3]{(x + 5)} - 3$.

Partner 2

Find the domain and range of $h(x) = -\sqrt{x + 1} + 4$.